

L7a: SIM Portfolios and a Risk-Free Asset

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Objectives for Today

By the end of this lecture, you should be able to:

- **Compare SIM and data-driven portfolios:** recall the inputs, identify omitted co-movement, and compare allocations under one covariance and on held-out prices.
- **Construct the capital allocation line and tangent portfolio:** derive the line relating expected growth to risk, identify its slope as the Sharpe ratio, and find the risky portfolio that maximizes it.
- **Choose and evaluate a complete portfolio:** explain lending, borrowing, and financing limits, and compare complete portfolios with an index fund.

Lectures and Examples

Lecture: SIM Portfolios and a Risk-Free Asset

Example 1: Compare Data-Driven and SIM Portfolios

Compare covariances, frontiers, and weights, then test on 2025 prices.

Example 2: Construct the Tangent Portfolio and Capital Allocation Line

Does every target pick the same risky portfolio? Test lending and borrowing in 2025.

Both examples run on the firms chosen in the L6b client interview, or on thirteen default firms. The L6b examples show how we estimated the SIM parameters.

Concept Review: The SIM Covariance

In the SIM $g_i = \alpha_i + \beta_i g_M + \varepsilon_i$, residuals have zero mean and are uncorrelated with the market and across assets, so $\mathbb{E}[g_i] = \alpha_i + \beta_i \mathbb{E}[g_M]$. **On the diagonal:**

$$\text{Var}(g_i) = \beta_i^2 \text{Var}(g_M) + 2\beta_i \underbrace{\text{Cov}(g_M, \varepsilon_i)}_{=0} + \text{Var}(\varepsilon_i) \quad \text{expand the SIM}$$

$$= \beta_i^2 \sigma_{g,M}^2 + \sigma_{g,\varepsilon,i}^2 \quad \text{second assumption}$$

Off the diagonal ($i \neq j$), the second and third assumptions remove three terms:

$$\begin{aligned} \text{Cov}(g_i, g_j) &= \beta_i \beta_j \text{Var}(g_M) + \beta_i \underbrace{\text{Cov}(g_M, \varepsilon_j)}_{=0} + \beta_j \underbrace{\text{Cov}(\varepsilon_i, g_M)}_{=0} + \underbrace{\text{Cov}(\varepsilon_i, \varepsilon_j)}_{=0} \\ &= \beta_i \beta_j \sigma_{g,M}^2 \end{aligned}$$

Concept Review: SIM Portfolio Inputs

With the betas in β , split each entry into its market and residual parts:

$$\Sigma_g = \sigma_{g,M}^2 \begin{bmatrix} \beta_1^2 & \beta_1\beta_2 & \cdots \\ \beta_2\beta_1 & \beta_2^2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} + \begin{bmatrix} \sigma_{g,\varepsilon,1}^2 & 0 & \cdots \\ 0 & \sigma_{g,\varepsilon,2}^2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} = \underbrace{\sigma_{g,M}^2 \beta\beta^\top}_{\text{market}} + \underbrace{\mathbf{D}_g}_{\text{residual}}$$

Substituting the training-data estimates gives the SIM inputs:

$$\hat{\mathbf{g}}_{\text{SIM}} = \hat{\alpha} + \hat{\beta} g'_M, \quad \hat{\Sigma}_{g,\text{SIM}} = s_{g,M}^2 \hat{\beta}\hat{\beta}^\top + \hat{\mathbf{D}}_g$$

- **Same means:** a least-squares line with an intercept passes through the sample means, so $\hat{\alpha}_i + \hat{\beta}_i g'_M = g'_i$.
- **Different covariance:** off the diagonal, the SIM covariance is the sample covariance with the residual covariances set to zero.

Portfolio Risk under a SIM

For fixed weights $\mathbf{w}$, substituting the SIM entries of Σ_g into the portfolio variance gives:

$$\begin{aligned}\text{Var}(g_p) &= \sum_i \sum_j w_i w_j \text{Cov}(g_i, g_j) && \text{portfolio variance (L6a)} \\ &= \sum_i w_i^2 (\beta_i^2 \sigma_{g,M}^2 + \sigma_{g,\varepsilon,i}^2) + \sum_i \sum_{j \neq i} w_i w_j \beta_i \beta_j \sigma_{g,M}^2 && \text{SIM entries} \\ &= \sigma_{g,M}^2 \sum_i \sum_j (w_i \beta_i)(w_j \beta_j) + \sum_i w_i^2 \sigma_{g,\varepsilon,i}^2 && \text{collect market terms} \\ &= \sigma_{g,M}^2 \left(\sum_i w_i \beta_i \right)^2 + \sum_i w_i^2 \sigma_{g,\varepsilon,i}^2 && \text{a perfect square} \\ &= \underbrace{\beta_p^2 \sigma_{g,M}^2}_{\text{market risk}} + \underbrace{\sum_i w_i^2 \sigma_{g,\varepsilon,i}^2}_{\text{residual risk}} && \beta_p = \sum_i w_i \beta_i\end{aligned}$$

Market risk depends on $\mathbf{w}$ only through β_p . Even weights over many assets with uncorrelated, similar-size residuals drive residual risk toward zero.

The Long-Only Problem with SIM Inputs

SIM-1 is L6a's long-only problem with SIM inputs and a floor $g_\star$ (year⁻¹):

$\begin{aligned} \underset{\mathbf{w}}{\text{minimize}} \quad & \underbrace{s_{g,M}^2 (\hat{\boldsymbol{\beta}}^\top \mathbf{w})^2}_{\text{market risk}} + \underbrace{\sum_i s_{g,\varepsilon,i}^2 w_i^2}_{\text{residual risk}} \\ \text{subject to} \quad & \hat{\mathbf{g}}_{\text{SIM}}^\top \mathbf{w} \geq g_\star \\ & \mathbf{1}^\top \mathbf{w} = 1 \\ & 0 \leq w_i \leq 1, \quad i \in \mathcal{P} \end{aligned}$	$\begin{aligned} \text{minimize} \quad & \mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_{g,\text{SIM}} \mathbf{w} \\ & \text{meet or exceed the growth target} \\ & \text{weights sum to one} \\ & \text{exclude short positions.} \end{aligned}$
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Sweeping $g_\star$ up from the long-only GMV growth rate traces the efficient frontier. We compare both allocations under one covariance, then on 2025 prices.

Example 1: [Compare Data-Driven and SIM Portfolios](#)

Company Profile: BlackRock

BlackRock is one of the world's largest investment managers. Larry Fink and seven partners founded it in 1988 to manage fixed-income assets. Its 2009 purchase of Barclays Global Investors brought the iShares funds.

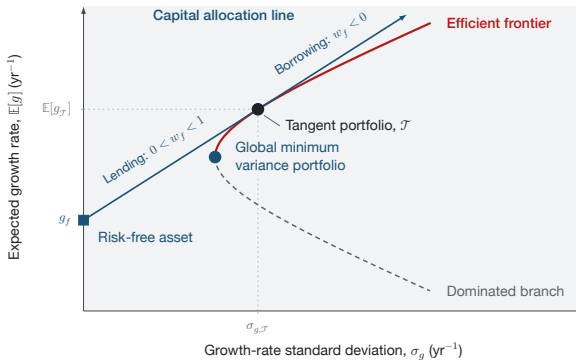
- **Index funds:** funds such as the iShares Core S&P 500 ETF hold a whole index.
- **Model portfolios:** Target Allocation portfolios suit different risk tolerances.
- **Portfolio risk:** Aladdin and Aladdin Risk analyze exposures and test scenarios.
- **Cash management:** money market funds and other short-term investments.

Today's connection: the tangent portfolio supplies the risky fund, and lending or borrowing sets the complete portfolio's risk.

Explore: [Student opportunities](#) [2027 Americas internship](#) [YouTube channel](#)

Adding a Risk-Free Asset

A **risk-free asset** has a known growth rate g_f (year^{-1}) and zero variance over the holding period, such as a Treasury bill held to maturity:



- **Capital allocation line (CAL):** mixing the risk-free asset with a risky portfolio gives a straight line from $(0, g_f)$ through that portfolio.
- **Tangent portfolio $\mathcal{T}$:** the steepest such line touches the efficient frontier at $\mathcal{T}$.
- **Lending and borrowing:** points before $\mathcal{T}$ lend at g_f , and points beyond it borrow at g_f .

The Capital Allocation Line

A **complete portfolio** c holds w_f in the risk-free asset and $1 - w_f$ in a risky portfolio p :

$$\mathbb{E}[g_c] = w_f g_f + (1 - w_f) \mathbb{E}[g_p] \quad \text{expectation is linear}$$

$$= g_f + (1 - w_f)(\mathbb{E}[g_p] - g_f) \quad \text{write } w_f = 1 - (1 - w_f)$$

$$\sigma_{g,c}^2 = (1 - w_f)^2 \sigma_{g,p}^2 \quad \text{the risk-free asset adds no variance}$$

$$\sigma_{g,c} = |1 - w_f| \sigma_{g,p} \quad \text{take the square root}$$

For $w_f \leq 1$ and $\sigma_{g,p} > 0$, substituting $1 - w_f = \sigma_{g,c}/\sigma_{g,p}$ gives the **CAL**:

$$\mathbb{E}[g_c] = g_f + \text{SR}_p \sigma_{g,c}, \quad \text{SR}_p = \frac{\mathbb{E}[g_p] - g_f}{\sigma_{g,p}} \text{ (Sharpe ratio)}$$

We lend for $0 < w_f < 1$, hold p alone for $w_f = 0$, and borrow for $w_f < 0$.

Daily data: the slope is a one-day Sharpe ratio. With independent increments, $\text{SR}_p^{\text{annual}} = \sqrt{252} \text{SR}_p$ (Sharpe, 1994).

The Tangent Portfolio

The **tangent portfolio** $\mathcal{T}$ maximizes the Sharpe ratio. Let Σ_g be positive definite, let the GMV portfolio's expected growth exceed g_f , and define $\mathbf{e} = \boldsymbol{\mu}_g - g_f \mathbf{1}$. With short positions allowed, solve:

$$\underset{\mathbf{w}}{\text{maximize}} \quad \frac{\mathbf{e}^\top \mathbf{w}}{\sqrt{\mathbf{w}^\top \Sigma_g \mathbf{w}}} \quad \text{subject to} \quad \mathbf{1}^\top \mathbf{w} = 1 \quad (\text{T-1})$$

At a stationary point, $\Sigma_g \mathbf{w} = (\mathbf{w}^\top \Sigma_g \mathbf{w} / \mathbf{e}^\top \mathbf{w}) \mathbf{e}$, so the weights point in the direction $\Sigma_g^{-1} \mathbf{e}$. Requiring them to sum to one gives:

$$\mathbf{w}_{\mathcal{T}} = \frac{\Sigma_g^{-1} (\boldsymbol{\mu}_g - g_f \mathbf{1})}{\mathbf{1}^\top \Sigma_g^{-1} (\boldsymbol{\mu}_g - g_f \mathbf{1})} \quad (\text{T-3})$$

With long-only weights, there is no closed form, so we solve numerically.

Two-Fund Separation

The steepest CAL separates choosing a risky portfolio from choosing how much risk to take. Mean-variance investors with the same inputs and risk-free rate hold the **same risky fund** $\mathcal{T}$, combined with the risk-free asset. Their risk preferences set w_f (Tobin, 1958).

Real investors face limits on how they finance their positions:

- **Long-only weights**: the risky fund is the long-only maximum-Sharpe portfolio, and investors still choose w_f .
- **No borrowing** ($w_f \geq 0$): the efficient boundary follows the CAL up to $\mathcal{T}$, then the risky efficient frontier.
- **Costlier borrowing**: a borrowing rate above g_f gives borrowers a flatter line through a different risky portfolio.

Risky and Risk-Free Allocation with SIM Inputs

Choose the risky weights $\mathbf{w}$ and put the rest, $w_f = 1 - \mathbf{1}^\top \mathbf{w}$, in the risk-free asset. For a target $g_\star > g_f$ (year⁻¹), problem SIM-2 is:

minimize $\mathbf{w}^\top \hat{\Sigma}_{g, \text{SIM}} \mathbf{w}$	minimize growth-rate variance
subject to $\hat{\mathbf{g}}_{\text{SIM}}^\top \mathbf{w} + \underbrace{(1 - \mathbf{1}^\top \mathbf{w})}_{w_f} g_f \geq g_\star$	meet or exceed the growth target
$0 \leq w_i \leq 1, \quad i \in \mathcal{P}$	exclude short positions.

The fraction w_f has no sign constraint, so $w_f < 0$ means borrowing at g_f .

Eliminating w_f gives the excess-growth constraint $(\hat{\mathbf{g}}_{\text{SIM}} - g_f \mathbf{1})^\top \mathbf{w} \geq g_\star - g_f$.

The Solutions Lie on a Ray

Scaling $\mathbf{w}$ by $c > 0$ scales the excess growth by c and the variance by c^2 . So, until a bound $w_i \leq 1$ binds, every target selects the same risky direction:

$$\mathbf{w}(g_\star) = (1 - w_f) \mathbf{w}_{\mathcal{T}}, \quad 1 - w_f = \frac{g_\star - g_f}{\hat{\mathbf{g}}_{\text{SIM}}^\top \mathbf{w}_{\mathcal{T}} - g_f} \quad (\text{SIM-3})$$

where the long-only tangent portfolio, $\mathbf{w}_{\mathcal{T}} = \mathbf{w} / \mathbf{1}^\top \mathbf{w}$, is any solution rescaled to sum to one.

- Targets below $\hat{\mathbf{g}}_{\text{SIM}}^\top \mathbf{w}_{\mathcal{T}}$ lend ($0 < w_f < 1$).
- Targets above it borrow ($w_f < 0$).

The estimation-risk notebook asks how much GMV and tangent allocations move when the estimated inputs change.

Example 2: [Construct the Tangent Portfolio and Capital Allocation Line](#)

Optional Advanced Material

The [advanced index](#) links the companions:

- [Tangent portfolio derivation](#): the full calculation behind T-1 to T-3 and the connection to risky and risk-free allocation.
- [Estimation risk in portfolio weights](#): resample trading days, compare GMV and tangent weights, and test long-only allocations on 2025 prices.

Summary

- **The covariance model changes the allocation:** the SIM keeps the sample means but sets the residual covariances to zero. Under the sample covariance, the SIM weights carry at least as much risk at every growth floor.
- **The tangent portfolio sets the steepest CAL:** the Sharpe ratio is the line's slope, and maximizing it selects the tangent portfolio.
- **Lending and borrowing set the complete portfolio's risk:** investors who can lend and borrow freely at the risk-free rate hold the same risky fund. Financing limits can change that fund.

The tangent portfolio supplies the risky fund, and lending or borrowing sets the complete portfolio's risk.